```
1000 'THIS SUBROUTINE IS CALLED FROM TABLE 20-4,  LINE 340
1010 '
1020 '  Variables entering subroutine:        PI, FC, LH, PR, HP, P%
1030 '  Variables exiting subroutine:         A0, A1, A2, B1, B2
1040 '  Variables used internally:            RP, IP, ES, VX, KX, T, W, M, D, K,
1050 '                                        X0, X1, X2, Y1, Y2
1060 '
1070 '                                        'Calculate the pole location on the unit circle
1080 RP = -COS(PI/(NP*2) + (P%-1) * PI/NP)
1090 IP =   SIN(PI/(NP*2) + (P%-1) * PI/NP)
1100 '
1110 '                                        'Warp from a circle to an ellipse
1120 IF PR = 0 THEN GOTO 1210
1130 ES = SQR( (100 / (100-PR))^2 -1 )
1140 VX = (1/NP) * LOG( (1/ES) + SQR( (1/ES^2) + 1) )
1150 KX = (1/NP) * LOG( (1/ES) + SQR( (1/ES^2) - 1) )
1160 KX = (EXP(KX) + EXP(-KX))/2
1170 RP = RP * ( (EXP(VX) - EXP(-VX) ) /2 ) / KX
1180 IP  = IP * ( (EXP(VX) + EXP(-VX) ) /2 ) / KX
1190 '
1200 '                                        's-domain to z-domain conversion
1210 T  = 2 * TAN(1/2)
1220 W  = 2*PI*FC
1230 M  = RP^2 + IP^2
1240 D = 4 - 4*RP*T + M*T^2
1250 X0 = T^2/D
1260 X1 = 2*T^2/D
1270 X2 = T^2/D
1280 Y1 = (8 - 2*M*T^2)/D
1290 Y2 = (-4 - 4*RP*T - M*T^2)/D
1300 '
1310 '                                        'LP TO LP, or LP TO HP transform
1320 IF LH = 1 THEN K = -COS(W/2 + 1/2) / COS(W/2 - 1/2)
1330 IF LH = 0 THEN K =  SIN(1/2 - W/2) / SIN(1/2 + W/2)
1340 D = 1 + Y1*K - Y2*K^2
1350 A0 = (X0 - X1*K + X2*K^2)/D
1360 A1 = (-2*X0*K + X1 + X1*K^2 - 2*X2*K)/D
1370 A2 = (X0*K^2 - X1*K + X2)/D
1380 B1 = (2*K + Y1 + Y1*K^2 - 2*Y2*K)/D
1390 B2 = (-(K^2) - Y1*K + Y2)/D
1400 IF LH = 1 THEN A1 = -A1
1410 IF LH = 1 THEN B1 = -B1
1420 '
1430 RETURN
```

TABLE 20-5

TABLE 20-4 and 20-5

Program to calculate the "a" and "b" coefficients for Chebyshev recursive filters. In lines 270-300, four parameters are entered into the program. The cutoff frequency, FC, is expressed as a fraction of the sampling frequency, and therefore must be in the range: 0 to 0.5. The variable, LH, is set to a value of *one* for a high-pass filter, and *zero* for a low-pass filter. The value entered for PR must be in the range of 0 to 29, corresponding to 0 to 29% ripple in the filter's frequency response. The number of poles in the filter, entered in the variable NP, must be an even integer between 2 and 20. At the completion of the program, the "a" and "b" coefficients are stored in the arrays A[ ] and B[ ] (a_0 = A[0], a_1 = A[1], etc.). TABLE 20-5 is a subroutine called from line 340 of the main program. Six variables are passed to this subroutine, and five variables are returned. Table 20-6 (next page) contains two sets of data to help debug this subroutine. The functions: COS and SIN, use radians, not degrees. The function: LOG is the natural (base *e*) logarithm. Declaring all floating point variables (including the value of π) to be double precision will allow more poles to be used. Tables 20-1 and 20-2 were generated with this program and can be used to test for proper operation. Chapter 33 describes the mathematical operation of this program.